14.10 FABRY-PEROT INTERFEROMETER

This instrument utilizes the fringes produced in the transmitted light after multiple reflection in the air film between two plane plates thinly silvered on the inner surfaces (Fig. 14L). Since the separation d between the reflecting surfaces is usually fairly large (from 0.1 to 10 cm) and observations are made near the normal direction, the fringes come under the class of fringes of equal inclination (Sec. 14.2). To observe the fringes, the light from a broad source (S_1S_2) of monochromatic light is allowed to traverse the interferometer plates E_1E_2. Since any ray incident on the first silvered surface is broken by reflection into a series of *parallel* transmitted rays, it is essential to use a lens L, which may be the lens of the eye, to bring these parallel rays together

* S. Tolansky, "Multiple-Beam Interferometry," Oxford University Press, New York, 1948.

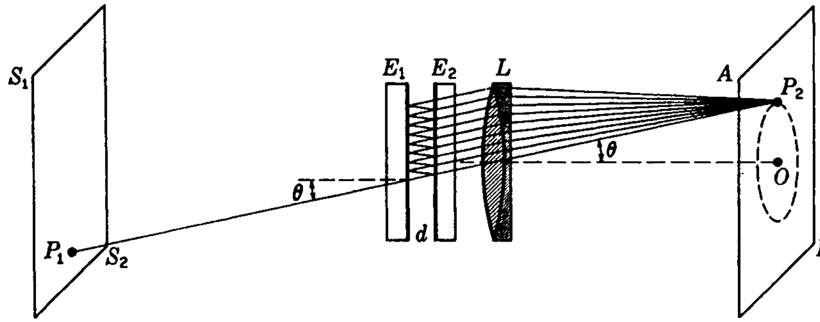

FIGURE 14L
Fabry-Perot interferometer E_1E_2 set up to show the formation of circular interference fringes from multiple reflections.

for interference. In Fig. 14L a ray from the point P_1 on the source is incident at the angle θ, producing a series of parallel rays at the same angle, which are brought together at the point P_2 on the screen AB. It is to be noted that P_2 is *not* an image of P_1. The condition for reinforcement of the transmitted rays is given by Eq. (14f) with $n = 1$ for air, and $\phi' = \theta$, so that

$$2d \cos \theta = m\lambda \qquad \textit{Maxima} \qquad (14p)$$

This condition will be fulfilled by all points on a circle through P_2 with their center at O, the intersection of the axis of the lens with the screen AB. When the angle θ is decreased, the cosine will increase until another maximum is reached for which m is greater by 1, 2, . . . , so that we have for the maxima a series of concentric rings on the screen with O as their center. Since Eq. (14p) is the same as Eq. (13g) for the Michelson interferometer, the spacing of the rings is the same as for the circular fringes in that instrument and they will change in the same way with change in the distance d. In the actual interferometer one plate is fixed, while the other may be moved toward or away from it on a carriage riding on accurately machined ways by a slow-motion screw.

14.11 BREWSTER'S* FRINGES

In a single Fabry-Perot interferometer it is not practicable to observe white-light fringes, since the condition of zero path difference occurs only when the two silvered surfaces are brought into direct contact. By the use of two interferometers in series, however, it is possible to obtain interference in white light, and the resulting fringes have had important applications. The two plane-parallel "air plates" are adjusted

* Sir David Brewster (1781–1868). Professor of physics at St. Andrew's, and later principal of the University of Edinburgh. Educated for the church, he became interested in light through repeating Newton's experiments on diffraction. He made important discoveries in double refraction and in spectrum analysis. Oddly enough, he opposed the wave theory of light in spite of the great advances in that theory made during his lifetime.

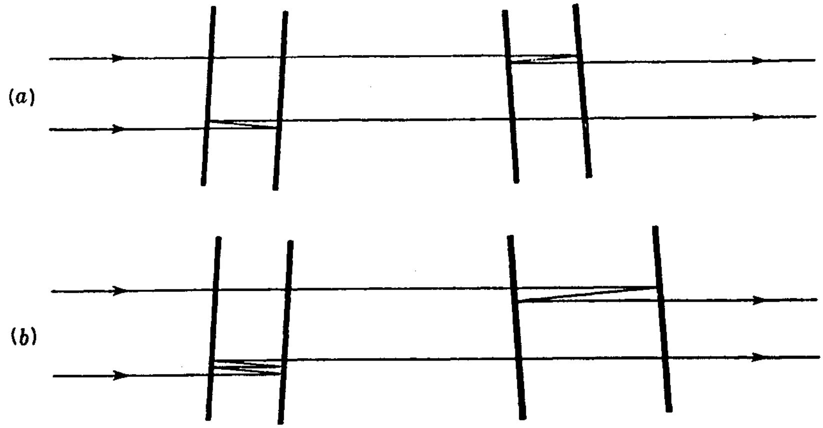

FIGURE 14M
Light paths for the formation of Brewster's fringes. (*a*) With two plates of equal thickness. (*b*) With one plate twice as thick as the other. The inclination of the two plates is exaggerated.

to exactly the same thickness, or else one to some exact multiple of the other, and the two interferometers are inclined to each other at an angle of 1 or 2°. A ray that bisects the angle between the normals to the two sets of plates can then be split into two, each of which after two or more reflections emerges, having traversed the same path. In Fig. 14M these two paths are drawn as separate for the sake of clarity, though actually the two interfering beams are derived from the same incident ray and are superimposed when they leave the system. The reader is referred to Fig. 13V, where the formation of Brewster's fringes by two thick glass plates in Jamin's interferometer is illustrated. A ray incident at any other angle than that mentioned above will give a path difference between the two emerging ones which increases with the angle, so that a system of straight fringes is produced.

The usefulness of Brewster's fringes lies chiefly in the fact that when they appear, the ratio of the two interferometer spacings is very exactly a whole number. Thus, in the redetermination of the length of the standard meter in terms of the wavelength of the red cadmium line, a series of interferometers was made, each having twice the length of the preceding, and these were intercompared using Brewster's fringes. The number of wavelengths in the longest, which was approximately 1 m long, could be found in a few hours by this method. It should finally be emphasized that this type of fringe results from the interference of only *two* beams and therefore cannot be made very narrow, as can the usual fringes due to multiple reflections.